

12 Joint Distributions

Example 12.1. Roll a fair four-sided die twice. Let X be the sum of the two dice, and let Y be the larger of the two rolls (or the common value if both rolls are the same). Recall Table 5.1.

1. Compute and interpret $p_{X,Y}(5,3) = P(X = 5, Y = 3)$.
2. Construct a “flat” table displaying the distribution of (X, Y) pairs, with one pair in each row.
3. Construct a two-way displaying the joint distribution on X and Y .
4. Sketch a plot depicting the joint distribution of X and Y .
5. Starting with the two-way table, how could you obtain $P(X = 5)$?

Table 12.1: Flat table representing the joint distribution of the sum (X) and larger (Y) of two rolls of a four-sided die.

(x, y)	$P(X = x, Y = y)$
(2, 1)	0.0625
(3, 2)	0.1250
(4, 2)	0.0625
(4, 3)	0.1250
(5, 3)	0.1250
(5, 4)	0.1250
(6, 3)	0.0625
(6, 4)	0.1250
(7, 4)	0.1250
(8, 4)	0.0625

6. Starting with the two-way table, how could you obtain the marginal distribution of X ? of Y ?

7. Starting with the marginal distribution of X and the marginal distribution of Y , could you necessarily construct the two-way table of the joint distribution? Explain.

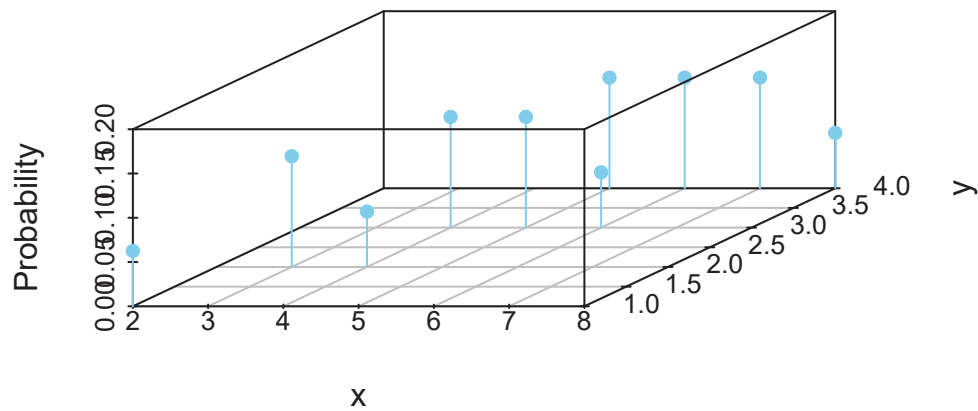
- The *joint distribution* of random variables X and Y is a probability distribution on (x, y) pairs, and describes how the values of X and Y vary together or jointly.
- Marginal distributions can be obtained from a joint distribution by “stacking”/“collapsing”/“aggregating” out the other variable.
- **In general, marginal distributions alone are not enough to determine a joint distribution.** (The exception is when random variables are *independent*.)

Table 12.2: Two-way table representing the joint distribution of the sum (X) and larger (Y) of two rolls of a four-sided die.

$x \setminus y$	1	2	3	4
2	1/16	0	0	0

3	0	2/16	0	0
4	0	1/16	2/16	0
5	0	0	2/16	2/16
6	0	0	1/16	2/16
7	0	0	0	2/16
8	0	0	0	1/16

3D Impulse Plot



Example 12.2. Continuing the dice rolling example, construct a spinner representing the joint distribution of X and Y .

- The **joint probability mass function (pmf)** of two *discrete* random variables (X, Y) defined on a probability space is the function defined by

$$p_{X,Y}(x, y) = P(X = x, Y = y) \quad \text{for all } x, y$$

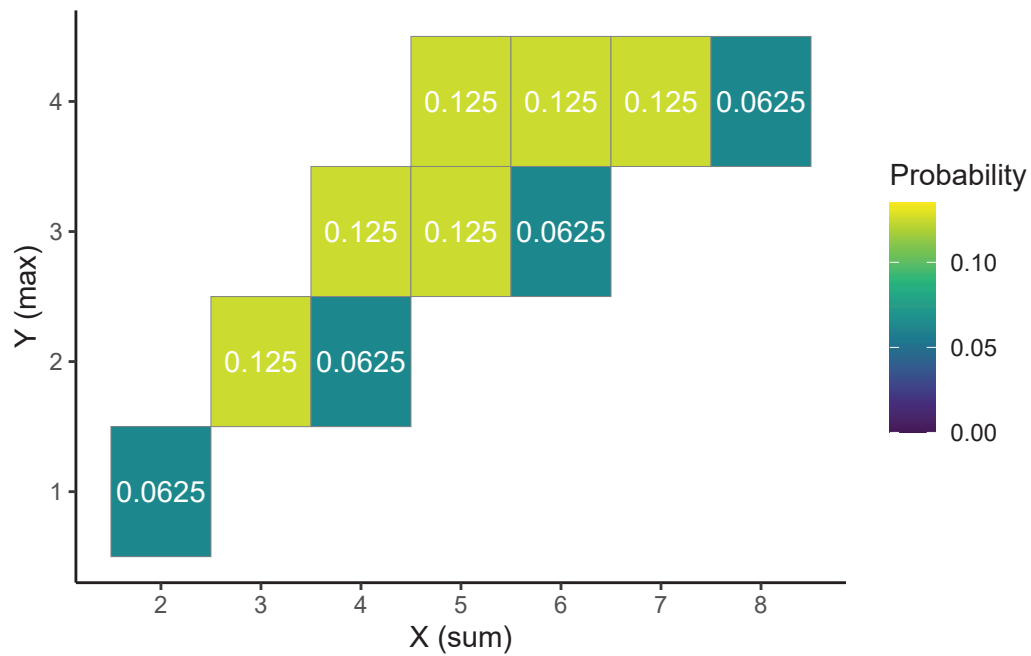


Figure 12.1: Tile plot representing the joint distribution of the sum (X) and larger (Y) of two rolls of a four-sided die.

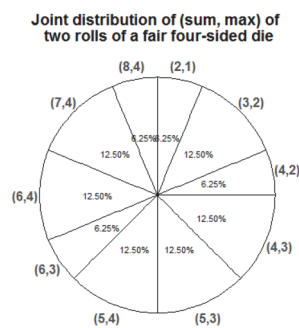


Figure 12.2: Spinner representing the joint distribution of the sum (X) and larger (Y) of two rolls of a fair four-sided die.

- Remember to specify the possible (x, y) pairs when defining a joint pmf.

Example 12.3. Let X be the number of home runs hit by the home team, and Y the number of home runs hit by the away team in a randomly selected Major League Baseball game. Suppose that X and Y have joint pmf

$$p_{X,Y}(x, y) = \begin{cases} e^{-2.3} \frac{1.2^x 1.1^y}{x!y!}, & x = 0, 1, 2, \dots; y = 0, 1, 2, \dots, \\ 0, & \text{otherwise.} \end{cases}$$

1. Compute and interpret the probability that the home teams hits 2 home runs and the away team hits 1 home run.
2. Construct a two-way table representation of the joint pmf (you can use software or a spreadsheet).
3. Compute and interpret the probability that each team hits at most 3 home runs.
4. Compute and interpret the probability that both teams combine to hit a total of 3 home runs.
5. Compute and interpret the probability that the home team and the away team hit the same number of home runs.

6. Compute and interpret the probability that the home team hits 2 home runs.

7. Compute and interpret the probability that the away team hits 1 home run.

- Recall that we can obtain marginal distributions from a joint distribution.
- *Marginal pmfs* are determined by the joint pmf via the law of total probability.
- If we imagine a plot with blocks whose heights represent the joint probabilities, the marginal probability of a particular value of one variable can be obtained by “stacking” all the blocks corresponding to that value.

$$p_X(x) = \sum_y p_{X,Y}(x, y) \quad \text{a function of } x \text{ only}$$
$$p_Y(y) = \sum_x p_{X,Y}(x, y) \quad \text{a function of } y \text{ only}$$

12.1 Joint probability density functions

- The joint distribution of two continuous random variables can be specified by a joint pdf, a *surface* specifying the density of (x, y) pairs.
- The probability that the (X, Y) pair of random variables lies in some region is the *volume* under the joint pdf surface over the region.
- The **joint probability density function (pdf)** of two *continuous* random variables (X, Y) defined on a probability space is the function $f_{X,Y}$ which satisfies, for any region

S

$$P[(X, Y) \in S] = \iint_S f_{X,Y}(x, y) dx dy$$

- A joint pdf is a surface with height $f_{X,Y}(x, y)$ at (x, y) .
- The probability that the (X, Y) pair of random variables lies in the region A is the *volume* under the pdf surface over the region A
- The height of the density surface at a particular (x, y) pair is related to the probability that (X, Y) takes a value “close to” (x, y) :

$$P(x - \epsilon/2 < X < x + \epsilon/2, y - \epsilon/2 < Y < y + \epsilon/2) = \epsilon^2 f_{X,Y}(x, y) \quad \text{for small } \epsilon$$

- The joint distribution is a distribution on (X, Y) pairs. A mathematical expression of a joint distribution is a function of both values of X and values of Y . Pay special attention to the possible values; the possible values of one variable might be restricted by the value of the other.
- The marginal distribution of Y is a distribution on Y values only, regardless of the value of X . A mathematical expression of a marginal distribution will have only values of the single variable in it; for example, an expression for the marginal distribution of Y will only have y in it (no x , not even in the possible values).

12.2 Independence of random variables

- Two random variables X and Y defined on a probability space with probability measure P are **independent** if $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$ for all x, y . That is, two random variables are independent if their joint cdf is the product of their marginal cdfs.
- Random variables X and Y are independent if and only if the joint distribution factors into the product of the marginal distributions. The definition is in terms of cdfs, but analogous statements are true for pmfs and pdfs.

Discrete RVs X and Y are independent

$$\iff p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for all } x, y$$

Continuous RVs X and Y are independent

$$\iff f_{X,Y}(x, y) = f_X(x)f_Y(y) \quad \text{for all } x, y$$

- Random variables X and Y are independent if and only if their joint distribution can be factored into the product of a function of values of X alone and a function of values of Y alone. That is, X and Y are independent if and only if there exist functions g and h for which

$$f_{X,Y}(x, y) \propto g(x)h(y) \quad \text{for all } x, y$$

- X and Y are independent if and only if the joint distribution factors into a product of the marginal distributions. The above result says that you can determine if that's true without first finding the marginal distributions.

Example 12.4. Let X be the number of home runs hit by the home team, and Y the number of home runs hit by the away team in a randomly selected Major League Baseball game. Suppose that X and Y are independent, X has a Poisson(1.2) distribution and Y has a Poisson(1.1) distribution.

1. Find the joint pmf of X and Y .
2. How could you have from the setup of Example 12.3 that X and Y were independent?

13 Covariance and Correlation

Quantities like expected value and variance summarize characteristics of the *marginal* distribution of a single random variable. When there are multiple random variables their *joint* distribution is of interest. Covariance and correlation summarize a characteristic of the joint distribution of two random variables, namely, the degree to which they “co-deviate from the their respective means”.

13.1 Covariance

- The **covariance** between two random variables X and Y is

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E(XY) - E(X)E(Y)\end{aligned}$$

- Covariance is the long run average of the products of the paired deviations from the mean
- Covariance is the expected value of the product minus the product of expected values.

Example 13.1. Roll a fair four-sided die twice and let X be the sum of the two rolls, and Y the larger of the two rolls.

1. Specify the joint pmf of X and Y . (These first few parts are review of things we have covered before.)
2. Find the marginal pmf of X and $E(X)$.
3. Find the marginal pmf of Y and $E(Y)$.

4. Find $E(XY)$. Is it equal to $E(X)E(Y)$?
5. Find $\text{Cov}(X, Y)$. Why is the covariance positive?
6. Describe how you would use simulation to approximate $\text{Cov}(X, Y)$.
- $\text{Cov}(X, Y) > 0$ (positive association): above average values of X tend to be associated with above average values of Y
 - $\text{Cov}(X, Y) < 0$ (negative association): above average values of X tend to be associated with below average values of Y
 - $\text{Cov}(X, Y) = 0$ indicates that the random variables are **uncorrelated**: there is no overall positive or negative association. But be careful: if X and Y are uncorrelated there can still be a relationship between X and Y ; there is just no overall positive or negative association.. We will see examples that demonstrate that being uncorrelated does not necessarily imply that random variables are independent.

Example 13.2. Continuing Example 13.1. Let X be the sum of the two rolls, Y the larger of the two rolls, W the number of rolls equal to 4, and Z the number of rolls equal to 1.

6. Is $\text{Cov}(X, W)$ positive, negative, or zero? Why?
7. Is $\text{Cov}(X, Z)$ positive, negative, or zero? Why?

8. Let $V = W + Z$. Is $\text{Cov}(X, V)$ positive, negative, or zero? Why?

9. Is $\text{Cov}(W, Z)$ positive, negative, or zero? Why?

Example 13.3. What is another name for $\text{Cov}(X, X)$?

13.2 Correlation

Example 13.4. The covariance between height in inches and weight in pounds for football players is 96.

1. What are the measurement units of the covariance?
2. Suppose height were measured in feet instead of inches. Would the shape of the joint distribution change? Would the strength of the association between height and weight change? Would the value of covariance change?

- The **correlation (coefficient)** between random variables X and Y is

$$\begin{aligned}\text{Corr}(X, Y) &= \text{Cov}\left(\frac{X - E(X)}{\text{SD}(X)}, \frac{Y - E(Y)}{\text{SD}(Y)}\right) \\ &= \frac{\text{Cov}(X, Y)}{\text{SD}(X)\text{SD}(Y)}\end{aligned}$$

- The correlation for two random variables is the covariance between the corresponding standardized random variables. Therefore, correlation is a *standardized* measure of the association between two random variables.
- A correlation coefficient has no units and is measured on a universal scale. Regardless of the original measurement units of the random variables X and Y

$$-1 \leq \text{Corr}(X, Y) \leq 1$$

- $\text{Corr}(X, Y) = 1$ if and only if $Y = aX + b$ for some $a > 0$
- $\text{Corr}(X, Y) = -1$ if and only if $Y = aX + b$ for some $a < 0$
- Therefore, correlation is a standardized measure of the strength of the *linear* association between two random variables.
- Covariance is the correlation times the product of the standard deviations.

$$\text{Cov}(X, Y) = \text{Corr}(X, Y)\text{SD}(X)\text{SD}(Y)$$

Example 13.5. The covariance between height in inches and weight in pounds for football players is 96. Heights have mean 74 and SD 3 inches. Weights have mean 250 pounds and SD 45 pounds. Find the correlation between weight and height.

Example 13.6. Consider the probability space corresponding to two rolls of a fair four-sided die. Let X be the sum of the two rolls, Y the larger of the two rolls. Compute $\text{Corr}(X, Y)$.

Example 13.7. Try the [Guessing Correlations applet!](#)

Example 13.8. Sketch a scatterplot where X and Y have a correlation of 0, but they are not independent.

14 Conditional Distributions

- The *joint distribution* of random variables X and Y is a probability distribution on (x, y) pairs, and describes how the values of X and Y vary together or jointly.
- We can also study *conditional distributions* of random variables given the values of some random variables. How does the distribution of Y change for different values of X (and vice versa)?

Example 14.1. Roll a fair four-sided die twice. Let X be the sum of the two rolls, and let Y be the larger of the two rolls (or the common value if a tie). We have previously found the joint and marginal distributions of X and Y , displayed in the two-way table below.

$p_{X,Y}(x, y)$					
$x \setminus y$	1	2	3	4	$p_X(x)$
2	1/16	0	0	0	1/16
3	0	2/16	0	0	2/16
4	0	1/16	2/16	0	3/16
5	0	0	2/16	2/16	4/16
6	0	0	1/16	2/16	3/16
7	0	0	0	2/16	2/16
8	0	0	0	1/16	1/16
$p_Y(y)$	1/16	3/16	5/16	7/16	

1. Compute $p_{X|Y}(6|4) = P(X = 6|Y = 4)$.
2. Construct a table, plot, and spinner to represent the conditional distribution of X given $Y = 4$.

3. Construct a table, plot, and spinner to represent the conditional distribution of X given $Y = 3$.
4. Construct a table, plot, and spinner to represent the conditional distribution of X given $Y = 2$.
5. Construct a table, plot, and spinner to represent the conditional distribution of X given $Y = 1$.
6. Compute $p_{Y|X}(4|6) = P(Y = 4|X = 6)$.
7. Construct a table, plot, and spinner to represent the distribution of Y given $X = 6$.
8. Construct a table, plot, and spinner to represent the distribution of Y given $X = 5$.
9. Construct a table, plot, and spinner to represent the distribution of Y given $X = 4$.

- The **conditional distribution of Y given $X = x$** is the distribution of Y values over only those outcomes for which $X = x$. It is a distribution on values of Y only; treat x as a fixed constant when conditioning on the event $\{X = x\}$.
- Conditional distributions can be obtained from a joint distribution by *slicing and renormalizing*. The conditional distribution of Y given $X = x$, where x represents a particular number, can be thought of as:
 - the slice of the joint distribution corresponding to $X = x$, a distribution on values of Y alone with $X = x$ fixed
 - renormalized so that the slice accounts for 100% of the probability over the values of Y
- The *shape* of the conditional distribution of Y given $X =$ is determined by the shape of the slice of the joint distribution over values of Y for the fixed x .
- For each fixed x , the conditional distribution of Y given $X = x$ is a different distribution on values of the random variable Y . There is not one “conditional distribution of Y given X ”, but rather a *family of conditional distributions of Y* given different values of X .
- Each conditional distribution is a distribution, so we can summarize its characteristics like mean and standard deviation. The conditional mean and standard deviation of Y given $X = x$ represent, respectively, the long run average and variability of values of Y over only (X, Y) pairs with $X = x$.
- Since each value of x typically corresponds to a different conditional distribution of Y given $X = x$, the conditional mean and standard deviation will typically be functions of x .

Example 14.2. We have already discussed two ways for simulating an (X, Y) pair in the dice rolling example: simulate a pair of rolls and measure X (sum) and Y (max), or spin the joint distribution spinner for (X, Y) once.

1. Now describe another way for simulating an (X, Y) pair using the spinners in Example 14.1. (Hint: you’ll need one more spinner in addition to the four from the previous example.)
2. Describe in detail how you can simulate (X, Y) pairs and use the results to approximate $P(X = 6|Y = 4)$.

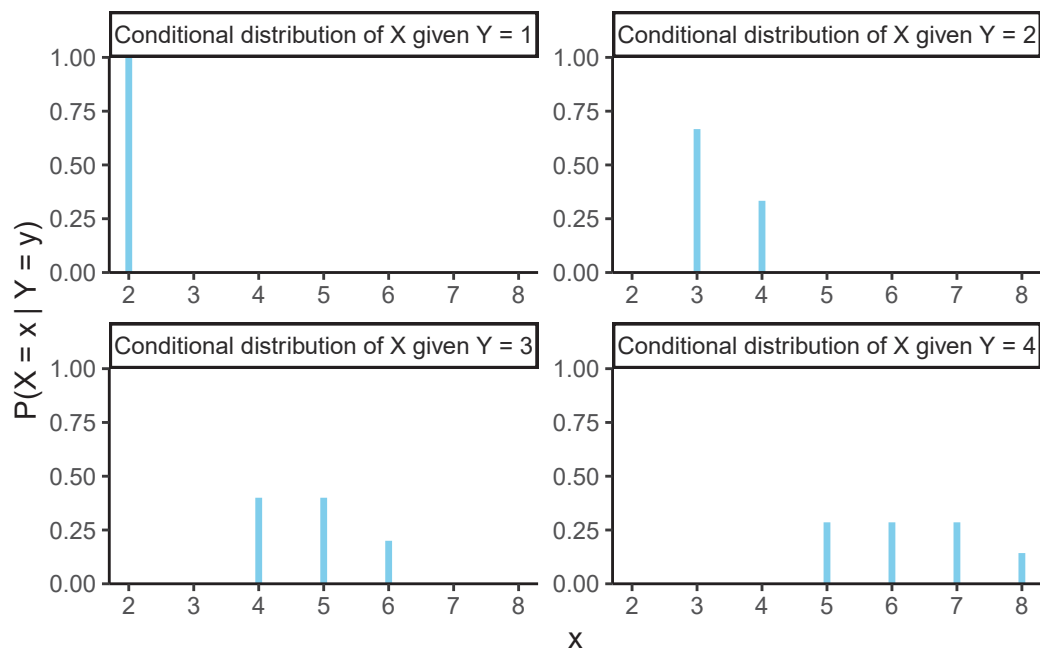
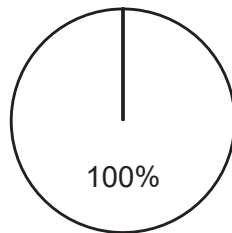
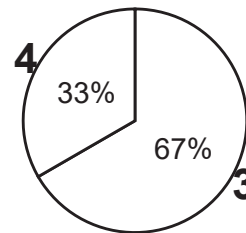


Figure 14.1: Impulse plots representing the family of conditional distributions of X given Y for the dice rolling example. Each plot represents a conditional distribution of X given $Y = y$ for a particular value of $y = 1, 2, 3, 4$.

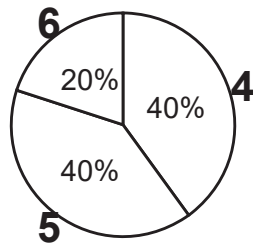
Distribution of X given $Y = 1$



Distribution of X given $Y = 2$



Distribution of X given $Y = 3$



Distribution of X given $Y = 4$

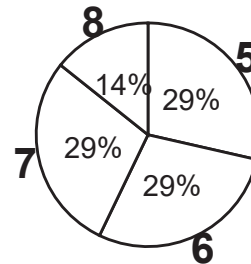


Figure 14.2: Spinners representing the family of conditional distributions of X given Y in the dice rolling example. Each spinner represents a conditional distribution of X given $Y = y$ for a particular value of $y = 1, 2, 3, 4$.

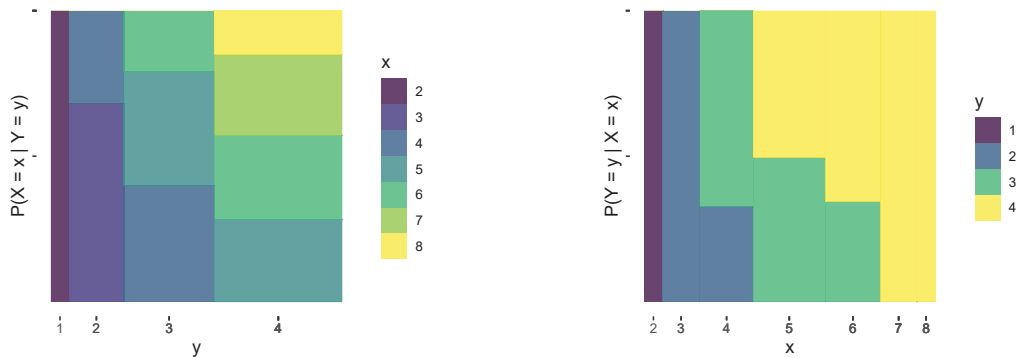
3. Describe in detail how you can simulate (X, Y) pairs and use the results to approximate the conditional distribution of X given $Y = 4$.

4. Describe in detail how you can simulate values from the conditional distribution of X given $Y = 4$ without simulating (X, Y) pairs.

- Rather than directly simulating from a joint distribution, we can simulate an (X, Y) pair in two stages:
 - Simulate a value of X from its marginal distribution. Call the simulated value x .
 - Given x , simulate a value of Y from the conditional distribution of Y given $X = x$. There will be a different distribution (spinner) for each possible value of x .
- This “marginal then conditional” process is essentially implementing the multiplication rule

$$\text{joint} = \text{conditional} \times \text{marginal}$$

- In many problems a joint distribution is naturally described by specifying the marginal distribution of X and the family of conditional distributions of Y given values of X
- Be sure to distinguish between joint, conditional, and marginal distributions.
- The joint distribution of X and Y is a distribution on (X, Y) pairs. A mathematical expression of a joint distribution is a function of both values of X and values of Y .
- The conditional distribution of Y given $X = x$ is a distribution on Y values (among (X, Y) pairs with a fixed value of $X = x$). A mathematical expression of a conditional distribution will involve both x and y , but x is treated like a fixed constant and y is treated as the variable. Note: the possible values of Y might depend on the value of x .
- The marginal distribution of Y is a distribution on Y values only, regardless of the value of X . A mathematical expression of a marginal distribution will have only values of the single variable in it; for example, an expression for the marginal distribution of Y will only have y in it (no x , not even in the possible values).



(a) Mosaic plot of conditional distributions of X given values of Y (a) Mosaic plot of conditional distributions of Y given values of X

- **Be careful** when conditioning with continuous random variables. Remember that the probability that a continuous random variable is equal to a particular value is 0; that is, for continuous X , $P(X = x) = 0$. - Mathematically, when we condition on $\{X = x\}$ we are really conditioning on $\{|X - x| < \epsilon\}$ — the event that the random variable X is within ϵ of the value x — and seeing what happens in the idealized limit when $\epsilon \rightarrow 0$.
- Practically, ϵ represents our “close enough” degree of precision, e.g., $\epsilon = 0.01$ if “within 0.01” is close enough.
- When conditioning on a continuous random variable X in a simulation, *never* condition on $\{X = x\}$; rather, condition on $\{|X - x| < \epsilon\}$ where ϵ represents the suitable degree of precision.

14.1 Conditional Expected Value

Example 14.3. Roll a fair four-sided die twice. Let X be the sum of the two rolls, and let Y be the larger of the two rolls (or the common value if a tie).

$p_{X,Y}(x,y)$					
$x \setminus y$	1	2	3	4	$p_X(x)$
2	1/16	0	0	0	1/16
3	0	2/16	0	0	2/16
4	0	1/16	2/16	0	3/16
5	0	0	2/16	2/16	4/16
6	0	0	1/16	2/16	3/16
7	0	0	0	2/16	2/16
8	0	0	0	1/16	1/16

$p_{X,Y}(x,y)$				
$p_Y(y)$	1/16	3/16	5/16	7/16

1. Compute and interpret $E(Y)$. How could you find a simulation-based approximation?
2. We have seen that the long run average value of Y is 3.125. Would you expect the conditional long run average value of Y given $X = 8$ to be greater than, less than, or equal to 3.125? Explain without doing any calculations. What about given $X = 3$?
3. How could you use simulation to approximate the conditional long run average value of Y given $X = 6$?
4. Compute and interpret $E(Y|X = 6)$.
5. Find $E(Y|X = x)$ for each possible value of x of X .
6. Compute and interpret $E(X|Y = 4)$. How could you find a simulation-based approximation?
7. Find $E(X|Y = y)$ for each possible value y of Y .

- The **conditional expected value** (a.k.a. *conditional expectation* a.k.a. *conditional mean*), of a random variable Y given the event $\{X = x\}$, defined on a probability space with measure P , is a *number* denoted $E(Y|X = x)$ representing the probability-weighted average value of Y , where the weights are determined by the conditional distribution of Y given $X = x$.

$$\text{Discrete } X, Y \text{ with conditional pmf } p_{Y|X}: \quad E(Y|X = x) = \sum_y y p_{Y|X}(y|x)$$

- Remember, when conditioning on $X = x$, x is treated as a fixed constant. The conditional expected value $E(Y|X = x)$ is a *number* representing the mean of the conditional distribution of Y given $X = x$.
- The conditional expected value $E(Y|X = x)$ is the long run average value of Y over only those outcomes for which $X = x$.
- To approximate $E(Y|X = x)$, simulate many (X, Y) pairs, discard the pairs for which $X \neq x$, and average the Y values for the pairs that remain.